

7.4 Belle Chess Hardware

Joe Condon and Ken Thompson

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Abstract

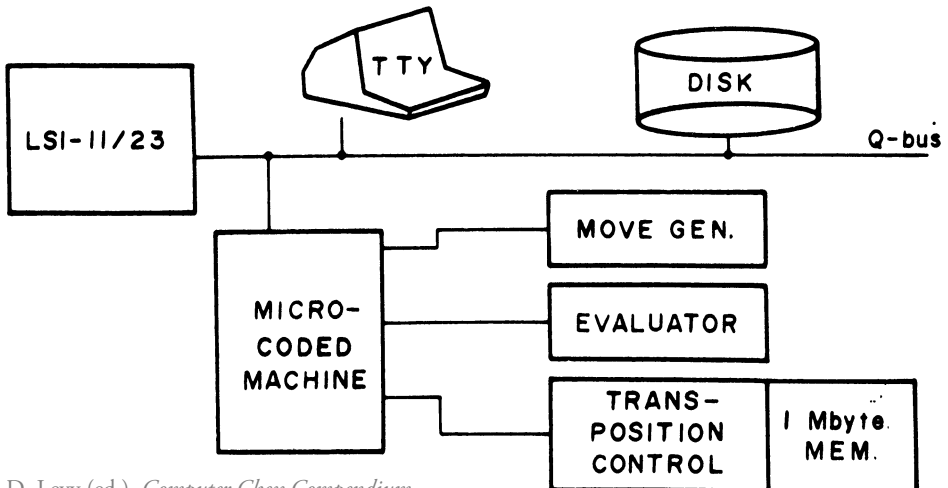
The computer chess program BELLE is currently the World Computer Chess Champion and the North American Computer Chess Champion. In human play, BELLE has consistently obtained master performance ratings. This paper describes the special-purpose hardware that gives BELLE its advantage: speed.

Introduction

The first version of BELLE was written in 1973 and first competed in the fourth ACM Championships in Atlanta. A slightly revised version competed the next year in San Diego. This experience made it painfully clear that computer chess belonged to the fastest computer. During this time, others were working on chess hardware. Moussouris, Holloway and Greenblatt (1979) constructed a highly publicized machine but failed to include an evaluation function. Despain designed a chess machine, described by Babaoglu (1977), but

never obtained funding for full construction. The authors built a small (25 chip) machine and entered it in the 1977 World Computer Chess Championships in Toronto. This first machine at best broke even on speed, but it whetted our appetites and gave us valuable experience. The next machine was quite a bit larger (325 chips). It had three separate sections - a move generator, a position evaluator and a transposition table. The alpha-beta search (Kunth and Moore, 1975) was still carried out by a minicomputer. This hardware/software combination proved to be comparable to the largest mainframes. It entered the 1978 (Washington DC) and 1979 (Detroit) ACM Championships, placing first and second respectively.

The latest machine was designed and constructed over a period of six months and contains approximately 1700 chips. The overall structure is the same as the previous machine, but the alpha-beta search has been embedded into hardware. The machine is controlled by an LSI-11/23



running a general purpose time-sharing system.

The LSI-11 contains the chess programs, opening book, and operating system software. It interfaces to a micro-coded alpha-beta machine that in turn controls a move generator, an evaluator and a transposition table memory.

The software version of BELLE examined chess positions at about 200 positions per second. The first chess machine did not change the speed much. The second machine increased the speed to about 5000 positions per second. The current machine runs at about 160,000 positions per second. These numbers are hard to compare, but the chess-specific computing power has increased about 800 times while the general purpose computing has decreased by a factor of 5.

Move Generator

The move generator is arranged in an 8×8 array of similar (except for edge effects) combinatorial circuits. The move generator performs two transformations. The transform to be performed is selected by a single wire *op-code*. The op-codes are named FIND-VICTIM and FIND-AGGRESSOR.

The FIND-VICTIM operation locates the highest valued piece under attack. The

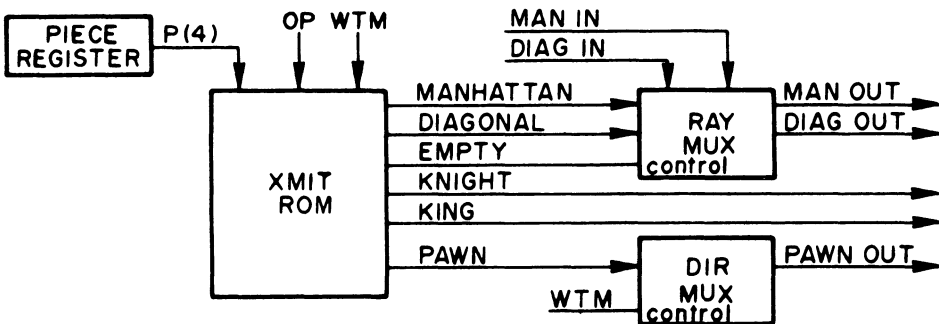
FIND-AGGRESSOR op-code finds the lowest valued piece attacking a given square. The given square is, of course, the previously found VICTIM square. The AGGRESSOR and the VICTIM squares are then the *from* and *to* squares for a chess move.

Transmitter

Each of the 64 circuits on the move generator consists of a transmitter, a receiver, a four-bit register holding the piece and a one-bit disable memory. Each transmitter sends signals to its (chess) neighbours. Likewise the receiver accepts signals from its neighbours.

The transmitter generates attacks radiating from a square. The main component of the transmitter is a read-only memory (ROM). The ROM generates a set of wires active if the square contains the corresponding piece. The global signal WHITE-TO-MOVE is true if White is on the move.

The rays (manhattan and diagonal) are propagated or generated at a square by a multiplexor selected with the square being empty or not. There are eight ray multiplexors on each square; one for each ray direction. The pawn attack outputs are multiplexed north or south selected by WHITE-TO-MOVE.



Receiver

The receiver section analyzes the attacks generated by the transmitters. The main component of the receiver is a programmed logic array (PLA).

The eight knight attacks are *ORed* together. Similarly the eight king attacks, the four diagonal (bishop) ray attacks, the four manhattan (rook) ray attacks, and the

two pawn attacks are *ORed* together. The PLA accepts the concentrated attack signals, the resident piece, WHITE-TO-MOVE, the disable bit and the op-code. It generates seven priority signals corresponding to the value of the attacked piece.

The king attacked is *ORed* over all 64 squares and is treated specially as a CHECK condition code for the micro-coded pro-